Marginal weights for the M13 particle-flow likelihood and score while preserving LGSSM and KSC-SV

BayesFilter technical note

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Implementation status, 6 October 2026. The marginal-mixture extension is implemented in commit `f5e69d716`. It accumulates the affine matrix and its tangent in the shared flow loop, evaluates both full Gaussian mixtures, and differentiates the finite recursion analytically. The original ancestor policy remains the default. The current main algorithm chapter includes the complete recursion, density and score derivations, and completed three-model evidence. This proposal preserves the original derivation and prospective optional defensive component; that component has not been implemented. References below to proposed experiments describe the October 4 plan. The October 5 experiments are complete: LGSSM/KSC showed material non-inferiority conditional on the tested datasets, and range–bearing showed lower errors than ancestor weights but still lost to bootstrap filtering on observed likelihood error. The expanded October 6 campaign tests fresh data, longer horizons, dimensions and cost; it does not itself run surrogate HMC.

The proposed repair is to sum over possible ancestors in the importance correction, while retaining the existing UKF covariance lifecycle, LEDH flow, Contract E reset, and analytical differentiation. This is the marginal particle-filter construction applied to the actual affine flow proposals. It changes the finite likelihood approximation and therefore needs a distinct route identity and its own tuning. It is not the unchanged Li–Coates Algorithm 1.

The purpose remains testing *surrogate HMC*. A particle-filter score may supply a force for proposals, but the Metropolis calculation must use the declared endpoint target. A better filter score can make those proposals more useful; it does not by itself establish posterior correctness or convergence.

The mathematics below establishes the proposed one-step integration identity and the total derivative of the resulting finite program. It also establishes why a universal finite-sample improvement claim would be wrong. The existing LGSSM and KSC-SV computations are preserved by retaining their existing route, then checking that preservation through the actual consumers. The new M13 route must earn admission by measurement.

1 The observed failure and the quantity to repair

The October 4 localization traced the M13 consumer through the fused batch route into the canonical scalar value and analytical-score implementation. At all eight diagnosed parameter points, the diagnostic trace and full consumer agreed in FP64. The analytical score also agreed with finite differences of its *own* implemented likelihood to about $2.6\,10^{-6}$ in the preceding score check. This rules out a large missing derivative term as the explanation supported by those checks. It does not establish accuracy relative to the statistical model.

M13 has a four-dimensional state and two observations (range and bearing). At the diagnosed low-process-noise point, the first observation already shows the failure:

Method	Log likelihood	First score coordinate	Particle ESS
Refined independent quadrature	−0.13791267	0.01149807	—
Same-design bootstrap	−0.13013616	0.00812633	205.04
Current LEDH	−2.22116858	3.57681741	6.66

There are 4,096 particles. The second score coordinates are respectively −0.10229746, −0.10443582, and 0.19335105. The diagnostic parameter is $\theta = (-0.73272085, -0.01465508)$ in the recorded M13 coordinates.

Before applying the observation density, the importance estimate of an integral equal to one is only 0.00344047. The first UKF position variance is about 116.74 times the conditional process variance. The root mean square flow displacement, measured in process-covariance Mahalanobis units, is 20.61. These observations support a specific mechanism: a child can be moved far from its own ancestor's narrow transition density. Its assigned ancestor then explains it poorly, even when other ancestors could explain the same location.

The discrepancy occurs before the first reset. Preserving the computed cloud's mean and covariance during reset cannot reconstruct filtering probability that the importance approximation has already lost. Replacing the UKF covariance with the process covariance is not justified by this observation: those covariances describe different conditional distributions, and the UKF lifecycle is part of the specified algorithm.

The quadrature was refined in both grid size and finite-difference step. This is a numerical reference, not an exact closed-form M13 score oracle. The eight points are now development data. Their results are descriptive and cannot serve as untouched promotion evidence.

2 Specify the finite mixture before changing its weights

Fix one time step and condition on the complete previous cloud and all offline numerical controls. Write the ancestors and their normalized positive weights as $(a_j, \omega_j)_{j=1}^N$, where $\sum_j \omega_j = 1$. For M13,

$$f_j(x) = p_\theta(x \mid a_j) = \mathcal{N}(x; m_j, Q_\theta), \qquad p_N(x) = \sum_{j=1}^N \omega_j f_j(x), \qquad Z_N = \int g_\theta(x) p_N(x)\, dx. \tag{1}$$

Here $m_j = Aa_j$ with the model's declared transition, and $g_\theta(x)$ is the implemented range–bearing observation density, including its angle convention. Z_N is the normalizer of this *finite predictive mixture*; it is not automatically the exact model's predictive likelihood.

In the current M13 implementation, each auxiliary flow starts at the zero-process-noise prediction. Conditional on the previous cloud, its coefficients do not depend on the current process innovation. The composed Euler map acting on that innovation's child is therefore affine. For a state dimension d (here $d = 4$), $u, x, c_j, m_j \in \mathbb{R}^d$ and $B_j, Q_\theta \in \mathbb{R}^{d\times d}$; assume Q_θ is symmetric positive definite and each B_j is nonsingular throughout the parameter neighborhood under consideration:

$$F_j(u) = B_j u + c_j, \qquad q_j(x) = \frac{f_j\left(B_j^{-1}(x - c_j)\right)}{|\det B_j|}. \tag{2}$$

These conditions make q_j a normalized pushforward density, as in the conditional-flow construction of [1, Algorithm 1 and Eqs. (19)–(20)]. Nonsingularity must hold in the neighborhood in which a score is claimed. For a general nonlinear map, a nonsingular Jacobian at the sampled point alone would not establish global invertibility; the conditional affine structure is doing real work here.

This construction uses a full-dimensional density with respect to Lebesgue measure. A structurally singular transition, such as a deterministic coordinate in AR(2), needs a separate innovation-space

measure derivation. The proposed full-rank route must reject that unsupported case, rather than add an undeclared ridge or call it the same density.

With one child $X_i = F_i(U_i)$ from each ancestor, $U_i \sim f_i$, the current correction has the form

$$V_i^{\text{old}} = \omega_i g_\theta(X_i) \frac{f_i(X_i)}{q_i(X_i)}. \tag{3}$$

Define the marginal proposal and implemented optional weights by

$$q_N(x) = \sum_j \omega_j q_j(x), \qquad V_i^{\text{mar}} = \omega_i g_\theta(X_i) \frac{p_N(X_i)}{q_N(X_i)}, \qquad \widehat{Z}_{\text{mar}} = \sum_i V_i^{\text{mar}}. \tag{4}$$

Both the numerator and denominator change to mixtures. Replacing only one of them computes a different correction.

The leading ω_i is correct for one draw from each component and the weighted sum displayed here. If instead M labels are drawn independently from ω, the estimator is the average of M marginal weights, with leading factor $1/M$, not a second factor ω_{J_i}. This distinction must survive any implementation refactor.

3 What marginalization proves, and what it does not

Proposition 1 (Conditional integration identity)**.** *Suppose each X_i has conditional marginal density q_i, $g_\theta p_N$ is integrable, and $q_N > 0$ wherever $g_\theta p_N > 0$. Then $\mathbb{E}[\widehat{Z}_{\text{mar}} \mid (a, \omega)] = Z_N$.*

Proof. Linearity of expectation and the definition of the mixture give

$$\begin{aligned} \mathbb{E}\widehat{Z}_{\text{mar}} &= \sum_i \omega_i \int q_i(x) \frac{g_\theta(x) p_N(x)}{q_N(x)}\, dx \\ &= \int \frac{\sum_i \omega_i q_i(x)}{q_N(x)} g_\theta(x) p_N(x)\, dx = Z_N. \end{aligned} \tag{5}$$

Independence between children is unnecessary for this expectation identity. □

Setting $g_\theta = 1$ gives the pre-observation normalization target one. This provides a cheap diagnostic of integration error. The statement is conditional on the specified cloud and sampling law. A frozen Halton trajectory is not a random experiment satisfying those assumptions merely because its inputs once came from a randomized construction. Neither this proof nor the marginal-particle-filter literature licenses a claim that the log likelihood, its score, or the complete likelihood product after the repository's deterministic Contract E and GenUT operations is unbiased. Unbiasedness of a positive likelihood estimator would not itself imply unbiasedness after taking its logarithm.

Proposition 2 (Variance reduction in the random-label experiment)**.** *Draw J with probabilities ω_j, then $X \mid J = j \sim q_j$. Let $W = g_\theta(X) f_J(X)/q_J(X)$ and $W_m = g_\theta(X) p_N(X)/q_N(X)$. If the relevant second moments are finite,* $\operatorname{Var}(W_m) \le \operatorname{Var}(W)$.

Proof. Bayes' rule gives $\Pr(J = j \mid X = x) = \omega_j q_j(x)/q_N(x)$. Hence

$$\mathbb{E}[W \mid X = x] = \sum_j \frac{\omega_j q_j(x)}{q_N(x)} \frac{g_\theta(x) f_j(x)}{q_j(x)} = \frac{g_\theta(x) p_N(x)}{q_N(x)} = W_m(x). \tag{6}$$

Applying the law of total variance,

$$\operatorname{Var}(W) - \operatorname{Var}(W_m) = \mathbb{E}[\operatorname{Var}(W \mid X)] \ge 0. \tag{7}$$

For two components, with $0 < w < 1$ and $q_1, q_2 > 0$, the pointwise second-moment difference has the explicit nonnegative form

$$\frac{wf_1^2}{q_1} + \frac{(1-w)f_2^2}{q_2} - \frac{(wf_1 + (1-w)f_2)^2}{wq_1 + (1-w)q_2} = \frac{w(1-w)(f_1q_2 - f_2q_1)^2}{q_1q_2(wq_1 + (1-w)q_2)}. \tag{8}$$

Multiply by g_θ^2 and integrate to obtain the corresponding second-moment comparison. □

This is the Rao–Blackwell argument behind the marginal particle filter [2, Sec. 3, Fig. 3, Sec. 3.1 and Prop. 1]. Its sampling experiment differs from a stratified allocation with exactly one child per component. The distinction prevents an invalid guarantee.

For an explicit counterexample, consider two states A, B with

$$g = (1, 2), \quad f_1 = (9/10, 1/10), \quad f_2 = (1/10, 9/10), \quad \omega_1 = \omega_2 = 1/2,$$

and choose $q_1 = (9/11, 2/11)$, $q_2 = (1/19, 18/19)$. Then $gf_1/q_1 = 11/10$ and $gf_2/q_2 = 19/10$, constant within each component. With independent draws, one from each component, the original estimate is exactly $3/2$ and has zero variance. But $p_N = (1/2, 1/2)$ and $q_N = (91/209, 118/209)$, so the marginal weights at A, B are $209/182$ and $209/118$. Direct calculation gives

$$\begin{aligned} \mathbb{E}\widehat{Z}_{\mathrm{mar}} &= 3/2, \\ \mathrm{Var}(\widehat{Z}_{\mathrm{mar}}) &= \frac{1}{4}\left(\frac{9}{11}\frac{2}{11} + \frac{1}{19}\frac{18}{19}\right)\left(\frac{209}{182} - \frac{209}{118}\right)^2 = \frac{555264}{28826161} > 0. \end{aligned} \tag{9}$$

Thus the proposed marginal weights can be worse under stratification. There is no universal finite-design non-deterioration theorem to transfer to LGSSM, KSC-SV, or M13. Even the valid random-label theorem says nothing by itself about the variance or accuracy of log-likelihood derivatives.

For M13, the reason to test the repair is narrower: the diagnosed ancestor-conditional ratios are extremely poorly integrated, whereas a mixture can explain a transported particle using other ancestors. Whether that opportunity is sufficient is an empirical question.

4 The score must differentiate the same finite program

Let D denote the total derivative in an arbitrary parameter direction, holding the base design and all offline controls fixed. It includes the dependence of previous particles, weights, current children, UKF moments, flow coefficients and process/observation covariances on the parameter. Work on a neighborhood where these functions are differentiable, densities are positive and the declared numerical status is valid. An angle-branch boundary or an invalid factorization requires an explicit status; it cannot be hidden in a smooth-score assertion.

At a child location $x = X_i(\theta)$ define the responsibilities

$$r_j^f = \frac{\omega_j f_j(x)}{p_N(x)}, \qquad r_j^q = \frac{\omega_j q_j(x)}{q_N(x)}.$$

Differentiating a finite sum and dividing by that sum yields

$$\begin{aligned} \mathrm{D}\log p_N(x) &= \sum_j r_j^f \{\mathrm{D}\log\omega_j + \mathrm{D}\log f_j(x)\}, \\ \mathrm{D}\log q_N(x) &= \sum_j r_j^q \{\mathrm{D}\log\omega_j + \mathrm{D}\log q_j(x)\}. \end{aligned} \tag{10}$$

For example, $\mathrm{D}p_N = \sum_j \{(\mathrm{D}\omega_j) f_j + \omega_j \mathrm{D}f_j\}$; division by p_N gives the first line. Responsibilities appear as coefficients in this first derivative; no extra derivative of the responsibility is added. That extra term would belong to a second derivative.

The proposed log-weight derivative is therefore

$$\mathrm{D}\log V_i^{\mathrm{mar}} = \mathrm{D}\log\omega_i + \mathrm{D}\log g_\theta(X_i) + \mathrm{D}\log p_N(X_i) - \mathrm{D}\log q_N(X_i). \tag{11}$$

With $\bar{V}_{ti} = V_{ti}/\sum_k V_{tk}$, the finite log likelihood and its derivative are

$$\widehat{\ell}(\theta) = \sum_t \log \sum_i V_{ti}(\theta), \qquad \mathrm{D}\widehat{\ell}(\theta) = \sum_t \sum_i \bar{V}_{ti} \mathrm{D}\log V_{ti}. \tag{12}$$

This is an ordinary finite-program chain rule, not differentiation under a probability integral. It proves which derivative an implementation must compute, rather than accuracy relative to the model likelihood.

To evaluate a proposal component at an arbitrary child, use the same d-dimensional domain: $B_j, \mathrm{D}B_j \in \mathbb{R}^{d\times d}$, with B_j nonsingular, and $x, c_j, v_j, \mathrm{D}x, \mathrm{D}c_j \in \mathbb{R}^d$. Put $v_j = B_j^{-1}(x - c_j)$. Differentiating $B_j v_j = x - c_j$ gives

$$\mathrm{D}v_j = B_j^{-1}\{\mathrm{D}x - \mathrm{D}c_j - (\mathrm{D}B_j)v_j\}. \tag{13}$$

On a nonsingular neighborhood the determinant has constant sign, and Jacobi's identity gives

$$\mathrm{D}\log q_j(x) = \mathrm{D}\log f_j(v_j) - \mathrm{tr}(B_j^{-1}\mathrm{D}B_j). \tag{14}$$

In particular, omitting the inverse-map tangent is wrong relative to the total-score claim.

For a Gaussian component take symmetric positive-definite $Q \in \mathbb{R}^{d\times d}$ and symmetric $\mathrm{D}Q \in \mathbb{R}^{d\times d}$. Let $x, m, e, u, \mathrm{D}x, \mathrm{D}m \in \mathbb{R}^d$, with $e = x - m$ and $u = Q^{-1}e$. From $\log f = -\frac{1}{2}\log\det Q - \frac{1}{2}e^\top Q^{-1}e + \text{const}$, $\mathrm{D}Q^{-1} = -Q^{-1}(\mathrm{D}Q)Q^{-1}$ and $\mathrm{D}\log\det Q = \mathrm{tr}(Q^{-1}\mathrm{D}Q)$, obtain

$$\mathrm{D}\log f(x) = u^\top(\mathrm{D}m - \mathrm{D}x) + \tfrac{1}{2}u^\top(\mathrm{D}Q)u - \tfrac{1}{2}\mathrm{tr}(Q^{-1}\mathrm{D}Q). \tag{15}$$

The terms in $\mathrm{D}x$, $\mathrm{D}m$ and $\mathrm{D}Q$ all matter. In the denominator, replace x by v_j and use (13).

The composed maps and their derivatives can be accumulated in the existing flow loop. For Euler step k let $M_k = I + \epsilon_k A_k$, with the step schedule fixed offline. Then

$$B_{k+1} = M_k B_k, \qquad c_{k+1} = M_k c_k + \epsilon_k b_k, \tag{16}$$

$$\mathrm{D}B_{k+1} = (\mathrm{D}M_k)B_k + M_k \mathrm{D}B_k, \qquad \mathrm{D}c_{k+1} = (\mathrm{D}M_k)c_k + M_k \mathrm{D}c_k + \epsilon_k \mathrm{D}b_k. \tag{17}$$

Initialize $B_0 = I$, $c_0 = 0$ and both tangents at zero. The existing analytical $A_k, b_k, \mathrm{D}A_k, \mathrm{D}b_k$ must be reused. Do not construct a simpler map in the batch lane or drop the UKF/auxiliary dependence. Matrix multiplication order must be preserved.

After weighting, the existing reset and correction still have to propagate the total derivative. In particular Contract E requires both the direct moment/weight terms and the terms through transport; changing the weights does not remove either dependence. The derivation is inductive: if the incoming cloud and UKF tangents are correct, the displayed formulas give correct weight tangents; the existing reset/correction must then return correct outgoing tangents. Wiring and end-to-end finite differences must check this induction. The implementation now follows this call chain. Fourteen full-filter FP64 finite-difference cases check both policies and every physical parameter direction in seven scopes at batch size one; broader scope checks are ongoing.

5 An optional defensive component if marginalization is insufficient

A mixture over flowed ancestors can still miss important regions. A second candidate, only if that problem persists, uses the actual proposal

$$r_j(x) = \alpha f_j(x) + (1-\alpha) q_j(x), \quad 0 < \alpha \le 1,$$
$$r_N(x) = \alpha p_N(x) + (1-\alpha) q_N(x), \qquad W_\alpha(x) = \frac{g_\theta(x) p_N(x)}{r_N(x)}. \tag{18}$$

Samples must be drawn from r_j, for example by an offline frozen, parameter-independent Bernoulli label and the corresponding base innovation. Merely changing the denominator while continuing to sample only from q_j is wrong. As with the earlier identity, the randomized sampling-law claim and any particular frozen realization are distinct. This proposal keeps α fixed; parameter-dependent or online selection would require another method and derivative derivation.

Since $r_N \ge \alpha p_N$ pointwise, wherever $p_N > 0$,

$$\frac{p_N}{r_N} \le \frac{1}{\alpha}, \qquad \mathbb{E}_{r_N} W_\alpha^2 = \int \frac{g_\theta^2 p_N^2}{r_N} \le \frac{1}{\alpha} \mathbb{E}_{p_N} g_\theta^2. \tag{19}$$

Provided $\mathbb{E}_{p_N} g_\theta^2 < \infty$, subtraction of Z_N^2 yields

$$\operatorname{Var}_{r_N}(W_\alpha) \le \frac{1}{\alpha} \operatorname{Var}_{p_N}(g_\theta) + \left(\frac{1}{\alpha} - 1\right) Z_N^2. \tag{20}$$

The last term must not be dropped. The bound is on a random marginal draw, not a uniform guarantee about stratified quadrature error, score error, or the adequacy of the previous cloud. It does not guarantee high ESS. At $\alpha = 1$ this is bootstrap sampling; at $\alpha = 0$ one obtains the unprotected marginal proposal, outside the displayed bound.

This is the basic defensive-importance argument discussed by Owen and Zhou [3, Sec. 3, especially Eq. (10), and Sec. 3.1]. Their stronger results use additional control variates; those are not part of this proposal and their guarantees must not be borrowed. A defensive mixture can deteriorate an already excellent importance proposal. No numerical value of α has been justified here. Its choice would need scope-specific offline calibration and a separate non-harm assessment, never selection inside HMC.

6 Preserve the original LGSSM and KSC-SV values and scores

Preservation has a precise and achievable meaning. Existing qualified scopes keep their existing algorithm, prepared observations, base designs, settings, coordinate transforms and consumer endpoints. The proposed correction is an explicit optional mode selected before tracing. It must not be a parameter-dependent switch and must not silently become the factory default. The mode is generic; this does not justify a reduced, model-specific implementation of the canonical flow.

Proposition 3 (Preservation by unchanged execution)**.** *Let $A_{\text{old}}(\theta; \xi, c)$ denote the current finite program with fixed inputs ξ and controls c. If the new factory's legacy mode calls that same program on those same inputs, with no changed operations, then on its valid differentiable domain*

$$A_{\text{legacy}}(\theta; \xi, c) = A_{\text{old}}(\theta; \xi, c), \qquad \mathrm{D}A_{\text{legacy}}(\theta; \xi, c) = \mathrm{D}A_{\text{old}}(\theta; \xi, c). \tag{21}$$

Proof. The function computed in legacy mode is the same function. Equal differentiable functions on a neighborhood have equal derivatives there. If the implementation returns a separate analytical score, preserving its operations and inputs also preserves that returned score, including any pre-existing error. This proposition does not newly certify that score. □

The implementation obligation is substantive: a refactor can change reduction order, random-number consumption, precision or a shared helper. The proof cannot certify those details before code exists. Freeze the current *post-fix* source closure and actual outputs, then exercise the real consumers to verify them. Bitwise parity is the goal where execution is deterministic. Any nondeterministic FP32 tolerance must come from measured same-build repeatability, not from a convenient new error allowance.

The original scalar LG1 model has state and observation dimensions one: $x_t = \phi x_{t-1} + \epsilon_t$ with process variance one, observation variance $1/2$, and stationary initial variance $(1-\phi^2)^{-1}$. Its exact Kalman value and analytical derivative are the independent reference. Preserve the derivative of the initial variance, $2\phi/(1-\phi^2)^2$, and the existing bounded-parameter chart and prior. The regression inventory must include every already qualified LGSSM consumer/scope, not infer coverage of all linear models from this one case.

The protected KSC consumer is the seven-component approximation to the log-squared-return observation law:

$$g_\theta(z \mid x) = \sum_{k=1}^{7} \pi_k \mathcal{N}(z; x + 2\log\beta + \mu_k, v_k). \tag{22}$$

It is distinct from the raw-return Gaussian stochastic-volatility observation density. For residual $e_k = z - x - 2\log\beta - \mu_k$ and fixed mixture constants, each component has

$$\mathrm{D}\log g_k = \frac{e_k}{v_k}\left(\mathrm{D}x + \frac{2\,\mathrm{D}\beta}{\beta}\right). \tag{23}$$

The recently repaired $2\,\mathrm{D}\beta/\beta$ term must remain. The score of (22) is the responsibility-weighted sum of these component scores. The $T = 2$ KSC fixture has an independent 49-component Gaussian likelihood after integrating its latent states; it is exact for this declared KSC mixture model, not for the unapproximated SV model. Preserve the stationary state variance and its derivative, chart, prior and observation mixture.

For both protected families, compare separately the physical likelihood, physical score, transformed target and transformed score. Check FP64 against the frozen post-fix implementation and the independent reference. Check the actual FP32 GPU/XLA consumer using the repository memory policy. Require the previous error envelope against the reference to be maintained for both values and scores; an improved value cannot excuse a worse score. Statuses, finiteness and deterministic outputs are part of preservation. Repeated builds or randomizations need paired uncertainty where relevant.

This proposal does not assert that marginal weighting will be as accurate as the old computation on every LGSSM or KSC scope. Migration would require new evidence. Retaining the original computation is what guarantees that testing the M13 candidate does not degrade those working results.

7 Implementation and the smallest discriminating experiments

The M13 consumer runs through `ledh_range_bearing_target_tf.py`, `ledh_canonical_batch_fused_tf.py`, and the shared executor `ledh_canonical_score_tf.py`. The optional policy `importance_weight_policy="marginal_mixture"` calls `marginal_prior_ratio_tangent` in `ledh_marginal_weights_tf.py`. The ancestor policy remains the default. Commit `f5e69d716` contains this implementation and its first completed three-model comparison.

The shared flow returns the affine matrix and its tangent together with the auxiliary endpoint and its tangent. The mixture routine evaluates both Gaussian mixtures at all children using log-sum-exp and analytical responsibility reductions. Automatic differentiation and finite differences are independent diagnostics only. At $N = 4096$ the exact chunk extent is $K = 2048$. There are N^2 component/child pairs per step: tiling controls memory, not this arithmetic cost. No nearest-ancestor truncation is used.

The completed 5 October comparison used untouched data and 32 paired particle designs. All 15 LGSSM/KSC parameter cells satisfied their declared 0.05 material non-inferiority criterion, conditional on the tested datasets. M13 value and score errors decreased relative to ancestor weighting in the four tested horizon/regime cells, but bootstrap filtering had smaller observed likelihood error in each cell. Thus this is an implemented partial repair, with evidence against promotion over the simple comparator. The main monograph's evidence section reports the numerical table. No defensive proposal component has been implemented or promoted.

The 6 October expanded master uses fresh datasets, starting at four and extending to eight or sixteen when uncertainty remains. It tests horizons $10, 20, 50, 120, 240$ and LGSSM dimensions $1, 3, 10, 20$, with native scalar KSC and four-state range–bearing models while the meaning of their dimension extensions is resolved. It also strengthens independent references, tests multiple-row parity and precision, and compares particle-count and runtime budgets with simple filters. Every scope freezes controls after separate calibration and validation. Its plan and running results are preserved in `docs/plans/surrogate-hmc-marginal-expanded-execution-2026-10-06.md`.

The expanded engineering checks have passed: multiple-row and separate-call values/scores were identical in all tested fixtures, and the largest relative finite-difference score discrepancy was 1.78×10^{-5}. Refined range–bearing $T = 1$ quadrature agreed with the replicated particle reference within its recorded uncertainty. Repeated $N = 4096$ full-filter precision checks were finite and repeatable; the largest FP32/FP64 log-likelihood difference was 0.00530, below the predeclared 0.05 materiality margin. These results do not settle the outstanding fresh-dataset replication grid.

Fresh-data runs have also finished for LGSSM $d = 1, T = 10, 20$ and $d = 3, T = 10$, at $N = 1024$: each used 16 independent datasets, 32 paired particle designs per dataset, and three parameter points. All nine cells met the 0.05 non-inferiority criterion for both primary metrics. Simultaneously adjusted approximate Student intervals supported smaller likelihood error in two cells and smaller scaled-score error in five cells. Other cells established no ranking. Dataset clusters, rather than individual particle designs, are the replication units; 16 clusters still give approximate interval coverage. These results do not establish an improvement over the exact Kalman filter.

Neither of the two tested control settings yielded valid calibration and validation throughout the $N = 1024$ scopes $d = 10, T = 10$, $d = 20, T = 10$, and $d = 20, T = 240$. Both weight policies encountered invalid states, so these scopes supply no accuracy comparison. Their failure is not yet localized. Larger particle counts and the outstanding nonlinear scopes remain in the master. Runtime comparisons now use FP32/TF32 simple filters against FP32/TF32 LEDH, with FP64 retained only for accuracy references. Because bootstrap filtering here returns a likelihood without a score, those timings cannot support a claim about HMC-force efficiency.

A subsequent surrogate-HMC test still needs the declared endpoint target, supported public tuner, adaptive warm-up, cumulative retained sampling and posterior/reference agreement. No new HMC samples are supplied by the component comparisons above. The defensive proposal remains a possible separately tuned repair if the fresh-data results diagnose inadequate coverage; its parameter α has not been selected.

The correct tuner must be chosen using `docs/reference/hmc-tuning-interface.md` and `HMC_TUNING_INTERFACE_CAPABILITIES`. The existing neural-force runner binding supports the public `tune_hmc_kernel` route; a stage helper is not a tuner, and an arbitrary force must not be routed through the fixed-transport tuner.

Every new scope still owes offline LEDH tuning. FP32 GPU/XLA remains the default execution target, with memory growth verified before device initialization; FP64 is a reference arm. No new tolerance is selected from the observed failures. For HMC materiality, useful explanatory quantities are local log-likelihood-ratio error $\delta(\theta + h) - \delta(\theta)$ and score error in the declared HMC coordinates, such as $\|L^\top(\widehat{s} - s_{\mathrm{ref}})\|$ for $\theta = \theta_0 + Lz$. Neither substitutes for downstream sampling evidence.

8 Mathematical audit and limits of this proposal

Manual proofs above establish the stated conditional identities. MathDevMCP has verified the two-component variance identity, conditional-expectation identity, rational mixture derivative, Gaussian scalar tangent, inverse-map scalar tangent, KSC tangent and exact counterexample arithmetic using its symbolic backend. Five additional coefficient checks cover the full two-dimensional Gaussian derivative; independent local algebra also checks noncommuting map composition and inverse differentiation. The separate audit record preserves exact inputs, outputs, parser limitations and document-audit results.

An attempted nested differentiation expression was not encoded correctly by the tool. It was replaced by an explicit rational derivative identity. A separate high-level request to refute universal stratified non-deterioration returned an inconclusive wrapper status despite an exact nonzero backend difference. The explicit counterexample and its checked rational arithmetic carry the argument; the wrapper abstention is not represented as a proof certificate.

These mathematical checks alone do not certify the TensorFlow call chain, empirical improvement, universal non-deterioration, a full-horizon exact M13 score or HMC convergence. The implementation and numerical evidence described above are separate from the symbolic proofs. The remaining empirical question is whether the optional correction improves accuracy and computational value across fresh datasets without material deterioration in LGSSM and KSC, and ultimately whether it supports the intended surrogate-HMC inference.

References

[1] Y. Li and M. Coates (2017). Particle Filtering with Invertible Particle Flow. *IEEE Transactions on Signal Processing*, 65(16), 4102–4116. Algorithm 1, Eqs. (19)–(20), Lemma IV.2, Theorem IV.3 and Lemma IV.4. https://arxiv.org/abs/1607.08799. The local implementation was inspected; the original-author software link was unavailable during this audit. No author-code equivalence claim follows from the paper inspection.

[2] M. Klaas, N. de Freitas and A. Doucet (2005). Toward Practical N^2 Monte Carlo: the Marginal Particle Filter. *Proceedings of UAI*, 308–315. Sec. 3 and Fig. 3 (mixture weights); Sec. 3.1 and Proposition 1 (random-label variance argument). https://www.cs.ubc.ca/~arnaud/klass_defreitas_doucet_marginalparticlefilterUAI2005.pdf.

[3] A. B. Owen and Y. Zhou (2000). Safe and Effective Importance Sampling. *Journal of the American Statistical Association*, 95(449), 135–143. Sec. 3, Eq. (10) and Sec. 3.1. https://artowen.su.domains/reports/seis.pdf.